[Total No. of Ques. 05] Roll No: [Total No. of Pages: 03]

G H Raisoni International Skill Tech University, Pune

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End Semester Examinations – Winter 2025

School of Engineering and Technology

F.Y. B.Tech., Semester - I (All)

Course Name: Matrices and Differential Calculus **Course Code:25UBSL1101**

Max Marks: 60 **Duration: 150 minutes**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any **TWO from A, B & C**. Question **D is compulsory**.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Find A^{-1} by using Adjoint method where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 3 & 2 & -1 \end{bmatrix}$	5	1	3
	B	For what valus of λ *and* μ the system of equation $x + y + z = 6$ $x + 2y + 3z = 10$ $x + 2y + \lambda z = \mu$ has (i) Unique solution (ii) No solution (iii) Infinite number of solutions	5	1	3
	C	Solve the system by inversion method. $x + 2y + 2z = 7$ $2x - 4y + z = -5$ $x + y + 2z = 5$	5	1	3
	D	Define Rank of Matrix. Find the Rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 1 & 2 \\ -1 & -2 & -3 & -1 & -2 \\ 0 & 2 & 2 & 3 & 2 \\ 2 & 4 & 6 & 2 & 4 \end{bmatrix}$	2	1	2

2		Solve any **TWO from Question A, B & C. Question D is compulsory.**			
	A	State and Verify Cayley Hamilton Theorem For $A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$ and hence find A^{-1} by using characteristic polynomials.	5	2	
	B	Find The Eigen Value and Eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$	5	2	3
	C	Find Eigen value and Eigen vectors of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 2 & 1 & 3 \end{bmatrix}$. Reduced the given matrix to diagonal form $(P^{-1}AP = D)$	5	2	3
	D	If 1,1,5 are the eigen values of $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$, find the eigen values of 5A and A^{-1}.	2	2	2
3		Solve any **TWO from Question A, B & C. Question D is compulsory.**			
	A	Using Taylor's theorem express $2x^3 + 7x^2 + x - 6$ in Powers of $(x - 2)$	5	3	3
	B	Verify Rolle's Theorem for the following function $f(x) = x^2(1 - x^2),\ x \in [0,1]$	5	3	3
	C	If $y = \sin^{-1}x$ then Prove that: $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$	5	3	3
	D	Find the nth derivative of $\sin 2x$	2	3	2
4		Solve any **TWO from Question A, B & C. Question D is compulsory.**			
	A	Discuss maxima and minima of $x^3y^3(1 - x - y)$	5	3	3
	B	If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$ show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (1 - 4\sin^2 u)\sin 2u$	5	3	3
	C	$If\ x = rsin\theta cos\emptyset,\ \ y = rsin\theta sin\emptyset,\ \ z = rcos\theta$ Show that find $\frac{\partial(x,y,z)}{\partial(r,\theta,\emptyset)} = r^2 sin\theta$	5	3	3
	D	IF $u = x^y$ find $\frac{\partial^2 u}{\partial x \partial y}$.	2	3	2
5		Solve any **TWO from Question A, B & C. Question D is compulsory.**			
	A	Find the directional derivative of the function $\emptyset = x\,y^2 + yz^3$ at a point $(1, -1, 1)$ along the vector $2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$	5	4	3

	B	Show that vectors $\bar{F} = (6xy + z^3)\hat{\imath} + (3x^2 - z)\hat{\jmath} + (3xz^2 - y)\hat{k}$ is irrotational. Also find the Scalar Potential Function ϕ such that $\bar{F} = \nabla\phi$	5	4	3
	C	If $\vec{F} = (x + 3y)\hat{\imath} + (y - 2z)\hat{\jmath} + (x + \lambda z)\hat{k}$ is solenoidal, then find the value of $\boldsymbol{\lambda}$.	5	4	3
	D	Given that $\bar{u} = xyz\,\hat{\imath} + (2x^2z - y^2x)\hat{\jmath} + xz^3\hat{k}$ and $\emptyset = xy + yz + z^2$. Find the divergence of $\bar{u}$ (i.e. $\nabla \cdot \bar{u}$) at $(1,0,-1)$	2	4	2

- End of the Question Paper -